

Concept 3 | Polynomial Theorems

English Student Edition • Focused formulas and guided practice

Formula opener

Remainder

$$P(a) = R \text{ when divisor} = x - a$$

Factor

$$P(a) = 0 \Rightarrow (x - a) \mid P(x)$$

Roots

$$P(x) = a \prod (x - r_i)$$

Classified Practice

Polynomial Theorems

Q001 Written

Find the remainder when $x^3 + 2x^2 - 3x - 4$ is divided by $x + 2$.

Q002 Written**Find the remainder when $3x^3 + 2x^2 - 5x + 6$ is divided by $x - 1$.**

Classified Practice

Polynomial Theorems

Q003 MCQ

Find the remainder when $x^3 + 2x^2 - 3x - 4$ is divided by $x + 2$ Select the correct answer:

Choose the correct option.

A $2 - 1$

B 2 (sign error)

C 2

D 3

Q004 MCQ**Find the remainder when $3x^3 + 2x^2 - 5x + 6$ is divided by $x - 1$ Select the correct answer:****Choose the correct option.****A** $6 - 1$ **B** 6 (sign error)**C** 6**D** 7

Classified Practice

Polynomial Theorems

Q005 Written

Find the remainder when $3x^4 - 8x^3 - 5x^2 + 12$ is divided by $3x - 2$.

Q006 Written**Find the remainder when $x^3 - 5x^2 + 4x - 1$ is divided by $2x - 1$.**

Classified Practice

Polynomial Theorems

Q007 Written

Find the remainder when $6x^3 - x^2 + x - 2$ is divided by $3x - 2$.

Q008 Written**Find the remainder when $4x^3 - 2x^2 - 4x + 3$ is divided by $2x + 1$.**

Classified Practice

Polynomial Theorems

Q009 MCQ

Find the remainder when $3x^4 - 8x^3 - 5x^2 + 12$ is divided by $3x - 2$ Select the correct answer:

Choose the correct option.

A 9

B $8 - 1$

C 8 (sign error)

D 8

Q010 MCQ**Find the remainder when $x^3 - 5x^2 + 4x - 1$ is divided by $2x - 1$ Select the correct answer:****Choose the correct option.**

A $-\frac{1}{8}$

B $-(-\frac{1}{8})$

C $-\frac{1}{8} + 1$

D $-\frac{1}{8} - 1$

Classified Practice

Polynomial Theorems

Q011 MCQ

Find the remainder when $6x^3 - x^2 + x - 2$ is divided by $3x - 2$ Select the correct answer:

Choose the correct option.

A 1

B $0 - 1$

C 0 (sign error)

D 0

Q012 MCQ**Find the remainder when $4x^3 - 2x^2 - 4x + 3$ is divided by $2x + 1$ Select the correct answer:****Choose the correct option.****A** 4**B** 5**C** $4 - 1$ **D** 4 (sign error)

Classified Practice

Polynomial Theorems

Q013 Written

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x + 3$ is 4, find a .

Q014 Written

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x - 3$ is 1, find a .

Classified Practice

Polynomial Theorems

Q015 Written

If the remainder when $x^3 - x^2 + ax - 4$ is divided by $x + 1$ is -2 , find a .

Q016 Written

If the remainder when $x^3 + 4x^2 - 5x + a$ is divided by $x + 3$ is 8, find a .

Classified Practice

Polynomial Theorems

Q017 MCQ

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x + 3$ is 4, find a Select the correct answer:

Choose the correct option.

A $a = 13/3$ (sign error)

B $a = 13/3$

C $a = 13/3 + 1$

D $a = 13/3 - 1$

Q018 MCQ

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x - 3$ is 1, find a Select the correct answer:

Choose the correct option.

A $a = -4$ (sign error)

B $a = -4$

C $a = -3$

D $a = -4 - 1$

Classified Practice

Polynomial Theorems

Q019 MCQ

If the remainder when $x^3 - x^2 + ax - 4$ is divided by $x + 1$ is -2 , find a Select the correct answer:

Choose the correct option.

A $a = -4 - 1$

B $a = -4$ (sign error)

C $a = -4$

D $a = -3$

Q020 MCQ

If the remainder when $x^3 + 4x^2 - 5x + a$ is divided by $x + 3$ is 8, find a Select the correct answer:

Choose the correct option.

A $a = -16$

B $a = -15$

C $a = -16 - 1$

D $a = -16$ (sign error)

Classified Practice

Polynomial Theorems

Q021 Written

If $x^3 - 3x^2 + 4x + a$ is divisible by $x - 2$, find a .

Q022 MCQ

If $x^3 - 3x^2 + 4x + a$ is divisible by $x - 2$, find a Select the correct answer:

Choose the correct option.

A $a = -4$

B $a = -3$

C $a = -4 - 1$

D $a = -4$ (sign error)

Classified Practice

Polynomial Theorems

Q023 Written

A polynomial $P(x)$ leaves remainders 3 on division by $x - 1$ and -5 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$.

Q024 Written

A polynomial $P(x)$ leaves remainders 3 on division by $x - 2$ and -7 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$.

Classified Practice

Polynomial Theorems

Q025 Written

A polynomial $P(x)$ leaves remainders 5 on division by $x - 1$ and -1 on division by $x + 2$. Find the remainder on division by $(x - 1)(x + 2)$.

Q026 Written

A polynomial $P(x)$ leaves remainders 4 on division by $x - 2$ and -11 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$.

Classified Practice

Polynomial Theorems

Q027 MCQ

A polynomial $P(x)$ leaves remainders 3 on division by $x - 1$ and -5 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$ Select the correct answer:

Choose the correct option.

A $2x + 1$

B $2x^2$

C $2x + 1 - 1$

D $2x + 1$ (sign error)

Q028 MCQ

A polynomial $P(x)$ leaves remainders 3 on division by $x - 2$ and -7 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$ Select the correct answer:

Choose the correct option.

A $-10x + 13$

B $-10x14$

C $-10x + 13 - 1$

D $-10x + 13$ (sign error)

Classified Practice

Polynomial Theorems

Q029 MCQ

A polynomial $P(x)$ leaves remainders 5 on division by $x - 1$ and -1 on division by $x + 2$. Find the remainder on division by $(x - 1)(x + 2)$ Select the correct answer:

Choose the correct option.

A $2x + 3$

B $2x^4$

C $2x + 3 - 1$

D $2x + 3$ (sign error)

Q030 MCQ

A polynomial $P(x)$ leaves remainders 4 on division by $x - 2$ and -11 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$ Select the correct answer:

Choose the correct option.

A $-15x + 19$

B $-15x20$

C $-15x + 19 - 1$

D $-15x + 19$ (sign error)

Classified Practice

Polynomial Theorems

Q031 Written

A polynomial $P(x)$ leaves remainder $7x + 6$ on division by $x^2 - x - 6$, and $4x - 3$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 + x - 2$.

Q032 Written

A polynomial $P(x)$ leaves remainder $3x + 2$ on division by $x^2 - x - 2$, and $7x - 6$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 - 1$.

Classified Practice

Polynomial Theorems

Q033 Written

A polynomial $P(x)$ leaves remainder $2x - 2$ on division by $x^2 - 2x - 3$, and $-x + 7$ on division by $x^2 - 4$. Find the remainder on division by $x^2 - x - 2$.

Q034 Written

A polynomial $P(x)$ leaves remainder $-x + 4$ on division by $x^2 - 3x + 2$, and $3x$ on division by $x^2 - 4x + 3$. Find the remainder on division by $x^2 - 5x + 6$.

Classified Practice

Polynomial Theorems

Q035 MCQ

A polynomial $P(x)$ leaves remainder $7x + 6$ on division by $x^2 - x - 6$, and $4x - 3$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 + x - 2$. Select the correct answer:

Choose the correct option.

A $3x - 2$ (sign error)

B $3x - 2$

C $3x - 1$

D $3x - 2 - 1$

Q036 MCQ

A polynomial $P(x)$ leaves remainder $3x + 2$ on division by $x^2 - x - 2$, and $7x - 6$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 - 1$ Select the correct answer:

Choose the correct option.

A 1

B x

C x (sign error)

D 0

Classified Practice

Polynomial Theorems

Q037 MCQ

A polynomial $P(x)$ leaves remainder $2x - 2$ on division by $x^2 - 2x - 3$, and $-x + 7$ on division by $x^2 - 4$. Find the remainder on division by $x^2 - x - 2$. Select the correct answer:

Choose the correct option.

- A $3x - 1$
- B $3x0$
- C $3x - 1 - 1$
- D $3x - 1$ (sign error)

Q038 MCQ

A polynomial $P(x)$ leaves remainder $-x + 4$ on division by $x^2 - 3x + 2$, and $3x$ on division by $x^2 - 4x + 3$. Find the remainder on division by $x^2 - 5x + 6$ Select the correct answer:

Choose the correct option.

- A $7x - 11$
 - B $7x - 12 - 1$
 - C $7x - 12$ (sign error)
 - D $7x - 12$
-
-
-

Classified Practice

Polynomial Theorems

Q039 Written

A polynomial $P(x)$ leaves remainder $x - 5$ on division by $(x + 1)^2$, and 6 on division by $x - 2$. Find the remainder on division by $(x + 1)^2(x - 2)$.

Q040 MCQ

A polynomial $P(x)$ leaves remainder $x - 5$ on division by $(x + 1)^2$, and 6 on division by $x - 2$. Find the remainder on division by $(x + 1)^2(x - 2)$ Select the correct answer:

Choose the correct option.

A $x^2 + 3x - 4$

B $x^2 + 3x - 3$

C $x^2 + 3x - 4 - 1$

D $x^2 + 3x - 4$ (sign error)

Classified Practice

Polynomial Theorems

Q041 **Written**

Using synthetic division, find the quotient and remainder when $x^3 + 5x + 11$ is divided by $x + 2$.

Q042 MCQ

Using synthetic division, find the quotient and remainder when $x^3 + 5x + 11$ is divided by $x + 2$ Select the correct answer:

Choose the correct option.

A $Q = x^2 - 2x + 9, R = -7$

B $Q = x^2 - 2x + 9, R = -6$

C $Q = x^2 - 2x + 9, R = -7 - 1$

D $Q = x^2 - 2x + 9, R = -7$ (sign error)

Classified Practice

Polynomial Theorems

Q043 Written

Using synthetic division, find the quotient and remainder when $A = x^3 + 5x - 6$ is divided by $B = x - 1$.

Q044 Written

Using synthetic division, find the quotient and remainder when $A = 3x^3 - 5x^2 - 3x - 1$ is divided by $B = x - 3$.

Classified Practice

Polynomial Theorems

Q045 **Written**

Using synthetic division, find the quotient and remainder when $A = x^3 - 5x + 6$ is divided by $B = x - 2$.

Q046 Written

Using synthetic division, find the quotient and remainder when $A = x^3 + 2x^2 - x - 3$ is divided by $B = x - 1$.

Classified Practice

Polynomial Theorems

Q047 Written

Using synthetic division, find the quotient and remainder when $A = 4x^2 - 2x + 2$ is divided by $B = x - 1$.

Q048 MCQ

Using synthetic division, find the quotient and remainder when $A = x^3 + 5x - 6$ is divided by $B = x - 1$ Select the correct answer:

Choose the correct option.

A $Q = x^2 + x + 6, R = 0$ (sign error)

B $Q = x^2 + x + 6, R = 0$

C $Q = x^2 + x + 6, R = 1$

D $Q = x^2 + x + 6, R = 0 - 1$

Classified Practice

Polynomial Theorems

Q049 MCQ

Using synthetic division, find the quotient and remainder when $A = 3x^3 - 5x^2 - 3x - 1$ is divided by $B = x - 3$. Select the correct answer:

Choose the correct option.

A $Q = 3x^2 + 4x + 9, R = 27$

B $Q = 3x^2 + 4x + 9, R = 26 - 1$

C $Q = 3x^2 + 4x + 9, R = 26$ (sign error)

D $Q = 3x^2 + 4x + 9, R = 26$

Q050 MCQ

Using synthetic division, find the quotient and remainder when $A = x^3 - 5x + 6$ is divided by $B = x - 2$ Select the correct answer:

Choose the correct option.

A $Q = x^2 + 2x - 1, R = 5$

B $Q = x^2 + 2x - 1, R = 4 - 1$

C $Q = x^2 + 2x - 1, R = 4$ (sign error)

D $Q = x^2 + 2x - 1, R = 4$

Classified Practice

Polynomial Theorems

Q051 MCQ

Using synthetic division, find the quotient and remainder when $A = x^3 + 2x^2 - x - 3$ is divided by $B = x - 1$. Select the correct answer:

Choose the correct option.

A $Q = x^2 + 3x + 2, R = -1$

B $Q = x^2 + 3x + 2, R = 0$

C $Q = x^2 + 3x + 2, R = -1 - 1$

D $Q = x^2 + 3x + 2, R = -1$ (sign error)

Q052 MCQ

Using synthetic division, find the quotient and remainder when $A = 4x^2 - 2x + 2$ is divided by $B = x - 1$ Select the correct answer:

Choose the correct option.

A $Q = 4x + 2, R = 4$ (sign error)

B $Q = 4x + 2, R = 4$

C $Q = 4x + 2, R = 5$

D $Q = 4x + 2, R = 4 - 1$

Classified Practice

Polynomial Theorems

Q053 Written

Using synthetic division, find the quotient and remainder when $A = 3x^4 + 5x^3 - 2x - 12$ is divided by $B = x + 2$.

Q054 MCQ

Using synthetic division, find the quotient and remainder when $A = 3x^4 + 5x^3 - 2x - 12$ is divided by $B = x + 2$ Select the correct answer:

Choose the correct option.

A $Q = 3x^3 - x^2 + 2x - 6, R = 1$

B $Q = 3x^3 - x^2 + 2x - 6, R = 0 - 1$

C $Q = 3x^3 - x^2 + 2x - 6, R = 0$ (sign error)

D $Q = 3x^3 - x^2 + 2x - 6, R = 0$

Classified Practice

Polynomial Theorems

Q055 Written

Factorise $x^3 + 2x^2 - x - 2$.

Q056 Written**Factorise** $x^3 - 2x^2 - 5x + 6$.

Classified Practice

Polynomial Theorems

Q057 Written

Factorise $x^3 - x^2 - 16x - 20$.

Q058 Written**Factorise** $2x^3 - x^2 - 5x - 2$.

Classified Practice

Polynomial Theorems

Q059 **Written****Factorise** $x^3 - 7x - 6$.

Q060 Written**Factorise** $x^3 + 4x^2 + x - 6$.

Classified Practice

Polynomial Theorems

Q061 Written

Factorise $x^3 + 6x^2 + 11x + 6$.

Q062 Written**Factorise** $x^3 - 3x + 2$.

Classified Practice

Polynomial Theorems

Q063 Written

Factorise $x^3 + x^2 - 8x - 12$.

Q064 Written**Factorise** $x^3 - 3x^2 - 9x - 5$.

Classified Practice

Polynomial Theorems

Q065 Written

Factorise $2x^3 - 3x^2 - 3x + 2$.

Q066 Written**Factorise** $3x^3 + x^2 - 8x + 4$.

Classified Practice

Polynomial Theorems

Q067 Written

Factorise $2x^3 + 7x^2 + 8x + 3$.

Q068 MCQ**Factorise $x^3 + 2x^2 - x - 2$ Select the correct answer:****Choose the correct option.**

A $(x - 1)(x + 1)(x^3)$

B $(x - 1)(x + 1)(x + 2) - 1$

C $(x - 1)(x + 1)(x + 2)$ (sign error)

D $(x - 1)(x + 1)(x + 2)$

Classified Practice

Polynomial Theorems

Q069 MCQ

Factorise $x^3 - 2x^2 - 5x + 6$ Select the correct answer:**Choose the correct option.**

- A $(x - 1)(x - 3)(x3)$
B $(x - 1)(x - 3)(x + 2) - 1$
C $(x - 1)(x - 3)(x + 2)$ (sign error)
D $(x - 1)(x - 3)(x + 2)$
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Q070 MCQ**Factorise $x^3 - x^2 - 16x - 20$ Select the correct answer:****Choose the correct option.**

- A $(x + 2)^2(x - 5) - 1$
 - B $(x + 2)^2(x - 5)$ (sign error)
 - C $(x + 2)^2(x - 5)$
 - D $(x + 2)^2(x - 4)$
-
-
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Classified Practice

Polynomial Theorems

Q071 MCQ

Factorise $2x^3 - x^2 - 5x - 2$ Select the correct answer:**Choose the correct option.**A $(x - 2)(2x + 1)(x + 1)$ (sign error)B $(x - 2)(2x + 1)(x + 1)$ C $(x - 2)(2x + 1)(x^2)$ D $(x - 2)(2x + 1)(x + 1) - 1$

Q072 MCQ

Factorise $x^3 - 7x - 6$ Select the correct answer:**Choose the correct option.**A $(x - 3)(x + 1)(x + 2)$ (sign error)B $(x - 3)(x + 1)(x + 2)$ C $(x - 3)(x + 1)(x3)$ D $(x - 3)(x + 1)(x + 2) - 1$

Classified Practice

Polynomial Theorems

Q073 MCQ

Factorise $x^3 + 4x^2 + x - 6$ Select the correct answer:**Choose the correct option.**

A $(x - 1)(x + 2)(x + 3) - 1$

B $(x - 1)(x + 2)(x + 3)$ (sign error)

C $(x - 1)(x + 2)(x + 3)$

D $(x - 1)(x + 2)(x^4)$

Q074 MCQ**Factorise $x^3 + 6x^2 + 11x + 6$ Select the correct answer:****Choose the correct option.****A** $(x + 1)(x + 2)(x + 3)$ (sign error)**B** $(x + 1)(x + 2)(x + 3)$ **C** $(x + 1)(x + 2)(x + 4)$ **D** $(x + 1)(x + 2)(x + 3) - 1$

Classified Practice

Polynomial Theorems

Q075 MCQ

Factorise $x^3 - 3x + 2$ Select the correct answer:**Choose the correct option.**

A $(x - 1)^2(x + 3)$

B $(x - 1)^2(x + 2) - 1$

C $(x - 1)^2(x + 2)$ (sign error)

D $(x - 1)^2(x + 2)$

Q076 MCQ

Factorise $x^3 + x^2 - 8x - 12$ Select the correct answer:**Choose the correct option.**

A $(x - 3)(x + 2)^3$

B $(x - 3)(x + 2)^2 - 1$

C $(x - 3)(x + 2)^2$ (sign error)

D $(x - 3)(x + 2)^2$

Classified Practice

Polynomial Theorems

Q077 MCQ

Factorise $x^3 - 3x^2 - 9x - 5$ Select the correct answer:**Choose the correct option.**

A $(x - 5)(x + 1)^3$

B $(x - 5)(x + 1)^2 - 1$

C $(x - 5)(x + 1)^2$ (sign error)

D $(x - 5)(x + 1)^2$

Q078 MCQ**Factorise $2x^3 - 3x^2 - 3x + 2$ Select the correct answer:****Choose the correct option.**

A $(x - 2)(x + 1)(2x - 1)$

B $(x - 2)(x + 1)(2x0)$

C $(x - 2)(x + 1)(2x - 1) - 1$

D $(x - 2)(x + 1)(2x - 1)$ (sign error)

Classified Practice

Polynomial Theorems

Q079 MCQ

Factorise $3x^3 + x^2 - 8x + 4$ Select the correct answer:**Choose the correct option.**

- A $(x - 1)(x + 2)(3x - 2) - 1$
B $(x - 1)(x + 2)(3x - 2)$ (sign error)
C $(x - 1)(x + 2)(3x - 2)$
D $(x - 1)(x + 2)(3x - 1)$
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Q080 MCQ**Factorise $2x^3 + 7x^2 + 8x + 3$ Select the correct answer:****Choose the correct option.**

A $(x + 1)^2(2x + 4)$

B $(x + 1)^2(2x + 3) - 1$

C $(x + 1)^2(2x + 3)$ (sign error)

D $(x + 1)^2(2x + 3)$

Classified Practice

Polynomial Theorems

Q081 **Written****Factorise** $x^4 - 6x^3 + 8x^2 + 8x - 16$.

Q082 Written**Factorise** $x^4 + 3x^3 - 27x^2 + 13x + 42$.

Classified Practice

Polynomial Theorems

Q083 Written

Factorise $x^4 - 4x^3 + 8x^2 - 8x + 3$.

Q084 Written**Factorise** $x^4 - 10x^3 + 35x^2 - 50x + 24$.

Classified Practice

Polynomial Theorems

Q085 **Written****Factorise** $x^4 - 5x^3 - 3x^2 + 17x - 10$.

Q086 Written**Factorise** $2x^4 - 5x^3 - 20x^2 + 20x + 48$.

Classified Practice

Polynomial Theorems

Q087 Written

Factorise $x^4 - 4x^3 - 4x^2 + 13x - 6$.

Q088 Written**Factorise** $x^4 - 4x^3 + 3x^2 + 2x - 6$.

Classified Practice

Polynomial Theorems

Q089 Written

Factorise $x^4 - x^3 + x^2 + x - 2$.

Q090 MCQ

Factorise $x^4 - 6x^3 + 8x^2 + 8x - 16$ Select the correct answer:**Choose the correct option.**

A $(x - 2)^2(x^2 - 2x - 3)$

B $(x - 2)^2(x^2 - 2x - 4) - 1$

C $(x - 2)^2(x^2 - 2x - 4)$ (sign error)

D $(x - 2)^2(x^2 - 2x - 4)$

Classified Practice

Polynomial Theorems

Q091 MCQ

Factorise $x^4 + 3x^3 - 27x^2 + 13x + 42$ Select the correct answer:**Choose the correct option.**

A $(x - 3)(x - 2)(x + 1)(x + 7)$

B $(x - 3)(x - 2)(x + 1)(x8)$

C $(x - 3)(x - 2)(x + 1)(x + 7) - 1$

D $(x - 3)(x - 2)(x + 1)(x + 7)$ (sign error)

Q092 MCQ

Factorise $x^4 - 4x^3 + 8x^2 - 8x + 3$ Select the correct answer:**Choose the correct option.**

A $(x - 1)^2(x^2 - 2x + 4)$

B $(x - 1)^2(x^2 - 2x + 3) - 1$

C $(x - 1)^2(x^2 - 2x + 3)$ (sign error)

D $(x - 1)^2(x^2 - 2x + 3)$

Classified Practice

Polynomial Theorems

Q093 MCQ

Factorise $x^4 - 10x^3 + 35x^2 - 50x + 24$ Select the correct answer:**Choose the correct option.**

A $(x - 4)(x - 3)(x - 2)(x - 1)$

B $(x - 4)(x - 3)(x - 2)(x - 1) - 1$

C $(x - 4)(x - 3)(x - 2)(x - 1)$ (sign error)

D $(x - 4)(x - 3)(x - 2)(x - 1)$

Q094 MCQ

Factorise $x^4 - 5x^3 - 3x^2 + 17x - 10$ Select the correct answer:**Choose the correct option.**

- A $(x - 5)(x - 1)^2(x + 2) - 1$
B $(x - 5)(x - 1)^2(x + 2)$ (sign error)
C $(x - 5)(x - 1)^2(x + 2)$
D $(x - 5)(x - 1)^2(x3)$
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Classified Practice

Polynomial Theorems

Q095 MCQ

Factorise $2x^4 - 5x^3 - 20x^2 + 20x + 48$ Select the correct answer:**Choose the correct option.**

A $(x - 4)(x - 2)(x + 2)(2x + 4)$

B $(x - 4)(x - 2)(x + 2)(2x + 3) - 1$

C $(x - 4)(x - 2)(x + 2)(2x + 3)$ (sign error)

D $(x - 4)(x - 2)(x + 2)(2x + 3)$

Q096 MCQ

Factorise $x^4 - 4x^3 - 4x^2 + 13x - 6$ Select the correct answer:**Choose the correct option.**

A $(x - 1)(x + 2)(x^2 - 5x + 3)$

B $(x - 1)(x + 2)(x^2 - 5x + 4)$

C $(x - 1)(x + 2)(x^2 - 5x + 3) - 1$

D $(x - 1)(x + 2)(x^2 - 5x + 3)$ (sign error)

Classified Practice

Polynomial Theorems

Q097 MCQ

Factorise $x^4 - 4x^3 + 3x^2 + 2x - 6$ Select the correct answer:**Choose the correct option.**A $(x - 3)(x + 1)(x^2 - 2x + 2)$ (sign error)B $(x - 3)(x + 1)(x^2 - 2x + 2)$ C $(x - 3)(x + 1)(x^2 - 2x3)$ D $(x - 3)(x + 1)(x^2 - 2x + 2) - 1$

Q098 MCQ

Factorise $x^4 - x^3 + x^2 + x - 2$ Select the correct answer:**Choose the correct option.**

A $(x - 1)(x + 1)(x^2 - x + 2) - 1$

B $(x - 1)(x + 1)(x^2 - x + 2)$ (sign error)

C $(x - 1)(x + 1)(x^2 - x + 2)$

D $(x - 1)(x + 1)(x^2 - x^3)$

Classified Practice

Polynomial Theorems

Q099 Written

Factorise $2x^3 - 3x^2 - x - 2$.

Q100 Written**Factorise** $2x^3 - 7x^2 + 3x + 6$.

Classified Practice

Polynomial Theorems

Q101 Written

Factorise $x^3 + 4x^2 + 2x - 4$.

Q102 Written**Factorise** $x^3 - 3x^2 + 2$.

Classified Practice

Polynomial Theorems

Q103 MCQ

Factorise $2x^3 - 3x^2 - x - 2$ Select the correct answer:**Choose the correct option.**

- A $(x - 2)(2x^2 + x + 1) - 1$
B $(x - 2)(2x^2 + x + 1)$ (sign error)
C $(x - 2)(2x^2 + x + 1)$
D $(x - 2)(2x^2 + x2)$
-
-
-

Q104 MCQ

Factorise $2x^3 - 7x^2 + 3x + 6$ Select the correct answer:

Choose the correct option.

A $(x - 2)(2x^2 - 3x - 3)$ (sign error)

B $(x - 2)(2x^2 - 3x - 3)$

C $(x - 2)(2x^2 - 3x - 2)$

D $(x - 2)(2x^2 - 3x - 3) - 1$

Classified Practice

Polynomial Theorems

Q105 MCQ

Factorise $x^3 + 4x^2 + 2x - 4$ Select the correct answer:**Choose the correct option.**A $(x + 2)(x^2 + 2x - 2)$ (sign error)B $(x + 2)(x^2 + 2x - 2)$ C $(x + 2)(x^2 + 2x - 1)$ D $(x + 2)(x^2 + 2x - 2) - 1$

Q106 MCQ**Factorise $x^3 - 3x^2 + 2$ Select the correct answer:****Choose the correct option.**

A $(x - 1)(x^2 - 2x - 1)$

B $(x - 1)(x^2 - 2x - 2) - 1$

C $(x - 1)(x^2 - 2x - 2)$ (sign error)

D $(x - 1)(x^2 - 2x - 2)$

Classified Practice

Polynomial Theorems

Q107 **Challenge****Evaluate the exact remainder when $x^{57} + 57$ is divided by $x + 1$.**

Classified Practice

Polynomial Theorems

Q108 Challenge

A model is $P(x) = 2x^3 - 3x^2 + ax + 6$. It must be divisible by $2x + 1$. Find a , then state what changes for divisor $x + 2$ and when more than one linear factor can divide a polynomial.

Classified Practice

Polynomial Theorems

Q109 Challenge

A polynomial $P(x)$ is divisible by $x - 1$ and leaves remainder -4 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$.

Classified Practice

Polynomial Theorems

Q110 Challenge

The polynomial $P(x) = 4x^4 - 4x^3 - 11x^2 + 12x - 3$ has a horizontal intercept at $x = 1/2$ with multiplicity 2. Find the other intercepts.

Classified Practice

Polynomial Theorems

Q111 Challenge

Let $P(x) = x^4 - 3x^3 + ax^2 + bx + 12$. If both $x - 1$ and $x + 2$ are factors, find a, b .

Classified Practice

Polynomial Theorems

Q112 Challenge

For the polynomial above with $a = -12$, $b = 2$, determine all roots of $P(x) = 0$.

Classified Practice

Polynomial Theorems

Q113 Challenge

Evaluate the exact remainder when $x^{57} + 57$ is divided by $x + 1$ Select the correct answer:

Choose the correct option.

A 56 (sign error)

B 56

C 57

D $56 - 1$

Classified Practice

Polynomial Theorems

Q114 Challenge

A model is $P(x) = 2x^3 - 3x^2 + ax + 6$. It must be divisible by $2x + 1$. Find a , then state what changes for divisor $x + 2$ and when more than one linear factor can divide a polynomial. Select the correct answer:

Choose the correct option.

- A $a = 10$; for $x + 2$, $a = -10$; linear factors correspond to roots.
- B $a = 10$; for $x + 2$, $a = -11$; linear factors correspond to roots (minus 1).
- C $a = 10$; for $x + 2$, $a = -11$; linear factors correspond to roots (sign error).
- D $a = 10$; for $x + 2$, $a = -11$; linear factors correspond to roots.

Classified Practice

Polynomial Theorems

Q115 Challenge

A polynomial $P(x)$ is divisible by $x - 1$ and leaves remainder -4 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$. Select the correct answer:

Choose the correct option.

- A $x - 1 - 1$
- B $x - 1$ (sign error)
- C $x - 1$
- D $x0$

Classified Practice

Polynomial Theorems

Q116 Challenge

The polynomial $P(x) = 4x^4 - 4x^3 - 11x^2 + 12x - 3$ has a horizontal intercept at $x = 1/2$ with multiplicity 2. Find the other intercepts Select the correct answer:

Choose the correct option.

A $x = \sqrt{3}, -\sqrt{3}$

B $x = \sqrt{3}, -\sqrt{4}$

C $x = \sqrt{3}, -\sqrt{3} - 1$

D $x = \sqrt{3}, -\sqrt{3}$ (sign error)

Classified Practice

Polynomial Theorems

Q117 Challenge

Let $P(x) = x^4 - 3x^3 + ax^2 + bx + 12$. If both $x - 1$ and $x + 2$ are factors, find a, b Select the correct answer:

Choose the correct option.

- A $a = -12, b = 2 - 1$
- B $a = -12, b = 2$ (sign error)
- C $a = -12, b = 2$
- D $a = -12, b = 3$

Classified Practice

Polynomial Theorems

Q118 Challenge

For the polynomial above with $a = -12$, $b = 2$, determine all roots of $P(x) = 0$ Select the correct answer:

Choose the correct option.

A $x = 1, -2, 2 \pm \sqrt{10}$

B $x = 1, -2, 2\sqrt{10}$

C $x = 1, -2, 2 - \sqrt{10}$

D $x = 1, -2, 2 + \sqrt{10}$

Classified Practice

Polynomial Theorems

Quiz MCQ 1 [Concept Quiz](#) · [MCQ](#)

Find the remainder when $P(x) = x^3 - 2x + 5$ is divided by $x - 2$ Select the correct answer:

Choose the correct option.

A $9 - 1$

B 9 (sign error)

C 9

D 10

Quiz MCQ 2 Concept Quiz · MCQ

If $P(x)$ is divisible by $x + 1$, which statement is true? Select the correct answer:

Choose the correct option.

- A $P(-1) = 0 - 1$
- B $P(-1) = 0$ (sign error)
- C $P(-1) = 0$
- D $P(-1) = 1$

Classified Practice

Polynomial Theorems

Quiz MCQ 3 [Concept Quiz](#) · MCQ

The remainder on division by a quadratic divisor has the form Select the correct answer:

Choose the correct option.

A 1

B $ax + b$

C $ax + b$ (sign error)

D 0

Quiz MCQ 4 Concept Quiz · MCQ

Which method records quotient and remainder for division by $x - a$? Select the correct answer:

Choose the correct option.

A 0

B 1

C *Syntheticdivision*

D *Syntheticdivision* (sign error)

Classified Practice

Polynomial Theorems

Quiz WRITTEN 5 Concept Quiz · Written

Find the remainder when $2x^3 + 3x^2 - x + 4$ is divided by $2x + 1$.

Quiz WRITTEN 6 Concept Quiz · Written

Find the remainder $R = ax + b$ when $P(x)$ has values $P(1) = 3$ and $P(-2) = -6$.

Classified Practice

Polynomial Theorems

Quiz WRITTEN 7 Concept Quiz · Written

Factorise $x^3 - 6x^2 + 11x - 6$.

Quiz WRITTEN 8 Concept Quiz · Written

If $x - 1$ and $x + 2$ are factors of $x^4 - 3x^3 + ax^2 + bx + 12$, find a, b .

Answer key

Use the question number shown on each page.

Q001 2	Q027 A	Q053 $Q = 3x^3 - x^2 + 2x - 6, R = 0$	Q079 C
Q002 6	Q028 A	Q054 D	Q080 D
Q003 C	Q029 A	Q055 $(x - 1)(x + 1)(x + 2)$	Q081 $(x - 2)(x^2 - 2x - 4)$
Q004 C	Q030 A	Q056 $(x - 1)(x - 3)(x + 2)$	Q082 $(x - 3)(x - 2)(x + 1)(x + 7)$
Q005 8	Q031 $3x - 2$	Q057 $(x + 2)^2(x - 5)$	Q083 $(x - 1)^2(x^2 - 2x + 3)$
Q006 $-\frac{1}{8}$	Q032 x	Q058 $(x - 2)(2x + 1)(x + 1)$	Q084 $(x - 4)(x - 3)(x - 2)(x - 1)$
Q007 0	Q033 $3x - 1$	Q059 $(x - 3)(x + 1)(x + 2)$	Q085 $(x - 5)(x - 1)^2(x + 2)$
Q008 4	Q034 $7x - 12$	Q060 $(x - 1)(x + 2)(x + 3)$	Q086 $(x - 4)(x - 2)(x + 2)(2x + 3)$
Q009 D	Q035 B	Q061 $(x + 1)(x + 2)(x + 3)$	Q087 $(x - 1)(x + 2)(x^2 - 5x + 3)$
Q010 A	Q036 B	Q062 $(x - 1)^2(x + 2)$	Q088 $(x - 3)(x + 1)(x^2 - 2x + 2)$
Q011 D	Q037 A	Q063 $(x - 3)(x + 2)^2$	Q089 $(x - 1)(x + 1)(x^2 - x + 2)$
Q012 A	Q038 D	Q064 $(x - 5)(x + 1)^2$	Q090 D
Q013 $a = 13/3$	Q039 $x^2 + 3x - 4$	Q065 $(x - 2)(x + 1)(2x - 1)$	Q091 A
Q014 $a = -4$	Q040 A	Q066 $(x - 1)(x + 2)(3x - 2)$	Q092 D
Q015 $a = -4$	Q041 $Q = x^2 - 2x + 9, R = -7$	Q067 $(x + 1)^2(2x + 3)$	Q093 D
Q016 $a = -16$	Q042 A	Q068 D	Q094 C
Q017 B	Q043 $Q = x^2 + x + 6, R = 0$	Q069 D	Q095 D
Q018 B	Q044 $Q = 3x^2 + 4x + 9, R = 26$	Q070 C	Q096 A
Q019 C	Q045 $Q = x^2 + 2x - 1, R = 4$	Q071 B	Q097 B
Q020 A	Q046 $Q = x^2 + 3x + 2, R = -1$	Q072 B	Q098 C
Q021 $a = -4$	Q047 $Q = 4x + 2, R = 4$	Q073 C	Q099 $(x - 2)(2x^2 + x + 1)$
Q022 A	Q048 B	Q074 B	Q100 $(x - 2)(2x^2 - 3x - 3)$
Q023 $2x + 1$	Q049 D	Q075 D	Q101 $(x + 2)(x^2 + 2x - 2)$
Q024 $-10x + 13$	Q050 D	Q076 D	Q102 $(x - 1)(x^2 - 2x - 2)$
Q025 $2x + 3$	Q051 A	Q077 D	Q103 C
Q026 $-15x + 19$	Q052 B	Q078 A	Q104 B

Q105 B**Q106** D**Q107** 56

$$a = 10 \quad \text{for } 2x + 1$$

Q108 $a = -11$ for $x + 2$ linear factors $\leftrightarrow$ roots**Q109** $x - 1$

Q110 $x = \sqrt{3}, -\sqrt{3}$

Q111 $a = -12, b = 2$

Q112 $x = 1, -2, 2 \pm \sqrt{10}$

Q113 B**Q114** D**Q115** C**Q116** A**Q117** C**Q118** A**Quiz MCQ 1** C**Quiz MCQ 2** C**Quiz MCQ 3** B**Quiz MCQ 4** C**Quiz WRITTEN 5** 4**Quiz WRITTEN 6** $R = 3x$ **Quiz WRITTEN 7** $(x - 1)(x - 2)(x - 3)$ **Quiz WRITTEN 8** $a = -12, b = 2$